ASSESSMENT PLAN

For

R Fundamental and Statistical Analysis for Beginners

TGS Ref No: TGS-2019504058

Conducted by
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Version 2.0

Table of Contents

1. Version Control Record

Version	Effective Date	Changes	Developer
1.0	11 Sep 2019	For Pilot Launch	Tertiary Infotech
1.1	9 Nov 2019	Minor Typo Correction	Kala Manickam
1.2	11 Dec 2019	Minor Typo Correction	Kala Manickam
1.3	2 Jun 2020	Modified the OQ	Dr. Alfred Ang
2.0	3 Jul 2024	● Update on the format ● Added the appeal procedure ● Added evidence collection and submission	Tertiary Infotech

2. Overview of Assessment

This assessment plan provides the essential guidelines for conducting the full assessment process for R Fundamental and Statistical Analysis for Beginners.

This assessment plan outlines the process of collecting evidence and making judgements on whether competency has been achieved. The technical principles of validity, reliability, flexibility and fairness are thus addressed in the conduct of the assessment, the development of the assessment tools and in the design, establishment and management of the assessment system

3. Context of Assessment

3.1. Skills Framework and TSC

The purpose of this assessment is to evaluate the candidate's performance against the following technical skills and competencies (TSC)

Skills Framework: Infocomm Technology
TSC Title: Analytics and Computational Modelling
TSC Code: ICT-DIT-3001-1.1

During the assessment, the candidate needs to show sufficient evidence through the stated assessment methods that he has the necessary knowledge and skills to perform all the abilities listed in the skills standard.

At the end of the assessment, the candidate's performance will be recorded in the assessment records. He will also be given feedback on his performance. Upon successful completion of his assessment, the candidate will be awarded a Statement of Attainment (SOA) for the competency

3.2. Candidate's Description

Demographics
- Typical age between 25 to 50 years, consisting of Singaporeans, Permanent Residence, Foreigners
- Varied Learners as some would have prior experience in IT industry while others are looking to gain knowledge to go into IT industry

Knowledge and Skills
- Able to operate using computer functions
- Minimum 3 GCE 'O' Levels Passes including English or WPL Level 5 (Average of Reading, Listening, Speaking & Writing Scores)

Attitude
- Positive Learning Attitude
- Enthusiastic Learner

Experience
- Minimum of 1 year of working experience.

4. Instructions to Assessor

4.1. Assessment Venue

The assessment is conducted in the classroom or training room

4.2. Assessment Methods

Learning Outcomes	Assessment Methods
LO1: Leaners will be able to install R and R Studio IDE K4 - Usage of analytics platforms and tools A2 - Use appropriate analytics platforms and analytical tools given specific analytics and reporting requirements	Oral Questioning (OQ) – K4, A2
LO2: Learners will be able to understand and code R data types K3 - Range and application of various types of data models K6 - Coding languages for programming of algorithms and signals K7 - Potential reasons for unintended outcomes A5 - Perform coding and configuration of software agents or programs based on a selected model or algorithm A9 - Implement changes to the coding and configuration of software	Written Assessment - Short-Answer Questions (WA-SAQ) – K3, K7 Case Study (CS) – K6, A5, A9
LO3: Learners will be able to understand and use R packages and datasets K3 - Range and application of various types of data models K6 - Coding languages for programming of algorithms and signals A2 - Use appropriate analytics platforms and analytical tools given specific analytics and reporting requirements A5 - Perform coding and configuration of software agents or programs based on a selected model or algorithm A9 - Implement changes to the coding and configuration of software	Written Assessment - Short-Answer Questions (WA-SAQ) – K3 Case Study (CS) – K6, A2, A5, A9

LO4: Learners will be able to perform data visualization in R K3 - Range and application of various types of data models K5 - Statistical modelling techniques A3 - Utilise a range of statistical methods and analytics approaches to data A4 - Conduct statistical modelling of data to derive patterns / solutions A7 - Diagnose unintended outcomes produced by analytical models A10 -Draw relevant trends and insights from data analysis to support decisions	Written Assessment - Short-Answer Questions (WA-SAQ) – K3, A3, A7 Case Study (CS) – K5, A4, A10
LO5: Learners will be able to understand and code R programming language K6 - Coding languages for programming of algorithms and signals K7 - Potential reasons for unintended outcomes A5 - Perform coding and configuration of software agents or programs based on a selected model or algorithm A7 - Diagnose unintended outcomes produced by analytical models A9 - Implement changes to the coding and configuration of software	Written Assessment - Short-Answer Questions (WA-SAQ) – K7, A7 Case Study (CS) – K6, A5, A9
LO6: Learners will be able to understand and use statistical methods in R K1 - Types of algorithms and advanced computational methods K2 - Range and application of various statistical algorithms K5 - Statistical modelling techniques A1 - Identify appropriate statistical algorithms and data models to test hypotheses or theories A3 - Utilise a range of statistical methods and analytics approaches to data A4 - Conduct statistical modelling of data to derive patterns / solutions A6 - Conduct tests on the actions taken and outcomes to assess effectiveness of the model A7 - Diagnose unintended outcomes produced by analytical models A8 - Propose changes or updates to the model or algorithms applied	Written Assessment - Short-Answer Questions (WA-SAQ) – K1, K2, A1, A3, A7 Case Study (CS) – K5, A4, A6, A8

4.3. Assessment Duration and Assessor to Candidate Ratio

Assessment Methods	Assessor to Candidate Ratio	Duration
WA-SAQ	1:3 (Min) 1:20 (Max)	1 hr
CS	1:3 (Min) 1:20 (Max)	80 mins
OQ	1:1	10 mins

4.4. Assessment Summary

Assessment Methods	Rationale
Written Assessment - Short-Answer Questions (WA-SAQ) Individual, Summative, Open book	Written Assessment - Short-Answer Questions (WA-SAQ) assessment will provide direct evidence of whether the candidates have acquired the underlying knowledge through written questions and answers. Written Assessment - Short-Answer Questions (WA-SAQ) assessment consists of open-ended questions that require candidates to create an answer to each question. It is effective for assessing the knowledge of a topic.
Case Study (CS) Individual, Summative, Open book	Case study (Written Assessment) consists of scenarios that assess the candidate's mastery of knowledge and abilities. It will enable consistent interpretation of evidence and reliable assessment outcomes. The assessor can collect ability and knowledge-based evidence to evaluate the candidate's competencies against the learning outcomes. This is a direct assessment method that will produce the product and knowledge-based evidence
Oral Questioning Individual, Summative	Oral questioning also provides another direct evidence that learners have acquired the underlying knowledge of R Fundamental and Statistical Analysis for Beginners. Assessors will ask direct questions to learners face to face.
Remarks: Oral Clarification will be conducted in a setting where the assessor-to-candidate ratio is 1:1 on a need basis to close minor performance gaps in the CS and WA-SAQ. Duration of up to 5 mins is allocated, which is not included in the assessment duration	

4.5. Evidence Collection and Submission

4.5.1. Written Assessment (Short Answer Questions)

File Submission: A hardcopy document should be submitted to the assessor. If necessary, a digital copy of the document can be submitted electronically via email or designated learning management system (LMS).

4.5.2. Case Study Assessment

File Submission: A hardcopy document should be submitted to the assessor. If necessary, a digital copy of the document can be submitted electronically via email or designated learning management system (LMS).

4.5.3. Oral Questioning

File Submission: Assessors will ask direct questions to learners face to face.

4.6. Appeal Procedures

The candidate has the right to challenge the assessment decision made by the Assessor. When giving feedback to the candidate about the assessment, the Assessor must ask the candidate if they agree with the outcome.

If the candidate agrees with the outcome, the Assessor and the candidate must sign the Assessment Outcome record. If the candidate does not agree with the outcome, the candidate should not sign the Assessment Outcome record.

If the candidate intends to appeal against the decision made, he/she should discuss the matter with the Assessor. If he/she is still not satisfied with the decision, the candidate must notify the Assessor of the intention to appeal. The Assessor will enter the intention in the Feedback section of the Assessment Outcome record.

The Assessor will notify training provider of the candidate's intention to lodge an appeal.

The Assessment Manager will collect information from the Assessor. A separate session will be arranged for the candidate and an independent Assessor appointed by training provider. An outcome agreed by all will conclude the appeal procedure.

A record of the appeal and any subsequent actions and findings will be submitted to SSG for record purposes.

5. Cross Reference Matrix of Assessment Methods

5.1. Knowledge Assessment Matrix

Knowledge	WA (SAQ)	CS	OQ
K1: Types of algorithms and advanced computational methods	√	√	-
K2: Range and application of various statistical algorithms	√	√	-
K3: Range and application of various types of data model	√	√	-
K4: Usage of analytics platforms and tools	-	-	√
K5: Statistical modelling techniques	√	√	-
K6: Coding languages for programming of algorithms and signals	√	√	-
K7: Potential reasons for unintended outcomes	√	√	-

5.2. Abilities Assessment Matrix

Ability	WA (SAQ)	CS	OQ
A1: Identify appropriate statistical algorithms and data models to test	√	√	-
A2: Use appropriate analytics platforms and analytical tools given specific analytics and reporting requirements	√	-	√
A3: Utilise a range of statistical methods and analytics approaches to data	√	√	-
A4: Conduct statistical modelling of data to derive patterns / solutions	√	√	-
A5: Perform coding and configuration of software agents or programs based on a selected model or algorithm	√	√	-
A6: Conduct tests on the actions taken and outcomes to assess effectiveness of the model	√	√	-
A7: Diagnose unintended outcomes produced by analytical models	√	√	-
A8: Propose changes or updates to the model or algorithms applied	√	√	-
A9: Implement changes to the coding and configuration of software	√	√	-
A10: Draw relevant trends and insights from data analysis to support decisions	√	√	-

5.3. Assessment Specifications

5.3.1. Assessment Specification for Written Assessment (SAQ)

Specifications	Written Assessment (SAQ)
Abilities and Knowledge	K1, K2, K3, K5, K6, K7 A1, A3, A4, A5, A6, A7, A8, A9, A10
Duration	1 hr
Venue	Training room / designated assessment area
Set up	**Physical Assessment (Main)** Candidates are required to sit in a manner whereby there is sufficient separation so that they will not be able to interact during the assessment. **Virtual Assessment** LMS ● Questions to be pre-loaded into LMS and opened only to the candidates and during the assessment time. ● Alternatively, an assessor will send the questions electronically to candidates only during the assessment time. Virtual Environment ● Learners to ensure the camera is on all the time and showing their full face. ● Learners to share their whole screen throughout the assessment period ● Assessors to ensure the recording is started before the candidates and remained recordings for every breakout room throughout all the duration of the Assessment.

Pre-assessment instructions for assessors	● Put the candidate at ease
	● Explain clearly the purpose and context of the assessment:
	  ● Purpose of the assessment
	  ● Assessment duration
	  ● Assessment methods and tools
	● Explain the methods to be used, the standards and criteria for the assessment and the duration of the assessment.
	● Explain the procedures of the assessment:
	  ● Check whether the candidate has any special requirements e.g. language, special needs, etc.
	  ● Ensure that the following documents are made available
	  ● Assessment Summary Record
	  ● Candidate's Appeal Form
	● Brief candidate on his/her rights and process of appeal
	● Identify any allowable adjustments that candidates may require should they have a special need
	
	In addition to the usual pre-assessment instructions, Include instructions on use of:
	  ● Code of Practice for Assessors (Annex A)

Pre-assessment instructions for candidates	● Be punctual ● Observe dress code: smart casual ● Request for assessor to repeat questions if question is not clear or if candidate has difficulty understanding assessor's accent ● To specify special needs, if any
Process of conducting assessment	● Verification of candidate’s identity ● Ensure candidates are seated properly so that they will not interact with each other ● Encourage the candidate to seek clarification if any ● Keep in mind the time limit for the assessment
Managing limitations of evidence	● Use of Oral Clarification
Recording Assessment Results	● Record the results in the Assessment Summary Record ● Check that all record forms, timing, dates, names, NRIC, signatures are completed. ● Check that all Assessment Criteria are ticked competent ● (C) or not yet competent (NYC) in the specific questions. As long as there is a NYC, the candidate is considered not successful in the entire assessment. The candidate is required to re-take the entire assessment. ● Record observation of Competence where applicable in ● the ‘Evidence of C and NYC must be recorded column ● Record reasons of Not Yet Competence where applicable in the ‘Evidence of C and NYC must be recorded column ● An ‘NYC’ in any K or A will results in an ‘NYC’ in the unit. ● A candidate is only deemed competent when they achieved a ‘C’ for all the A and K in the unit.
Feedback	● Give clear and constructive feedback to candidate using appropriate language ● Seek clarification and agreement with the candidate on the outcome ● Provide guidance on overcoming gaps in competency ● Ask the candidates to sign the assessment record ● Record the feedback on the assessment record ● Sign off the assessment record ● Assessor should reiterate the appeal process (if the candidate is assessed as not yet competent)

5.3.2. Assessment Specification for Case Study

Specifications	Case Study
Abilities and Knowledge	K1, K2, K3, K5, K6, K7 A1, A3, A4, A5, A6, A7, A8, A9, A10
Duration	80 mins
Venue	Training room / designated assessment area
Set up	**<u>Physical Assessment</u>** Candidates are required to sit in a manner whereby there is sufficient separation so that they will not be able to interact during the assessment. **<u>Online Assessment (Main)</u>** <u>LMS</u> ● Questions to be pre-loaded into LMS and opened only to the candidates and during the assessment time. ● Alternatively, the assessor will send the questions electronically to candidates only during the assessment time. <u>Virtual Environment</u> ● Learners to ensure the camera is on all the time and showing their full face. ● Learners to share their whole screen throughout the assessment period ● Assessors to ensure the recording is started before the candidates and remained recordings for every breakout room throughout all the duration of the Assessment.

Pre-assessment instructions for assessors	● Put the candidate at ease
	● Explain clearly the purpose and context of the assessment:
	  ● Purpose of the assessment
	  ● Assessment duration
	  ● Assessment methods and tools
	● Explain the methods to be used, the standards and criteria for the assessment and the duration of the assessment.
	● Explain the procedures of the assessment:
	  ● Check whether the candidate has any special requirements e.g. language, special needs, etc.
	  ● Ensure that the following documents are made available
	  ● Assessment Summary Record
	  ● Candidate's Appeal Form
	● Brief candidate on his/her rights and process of appeal
	● Identify any allowable adjustments that candidates may require should they have a special need
	
	In addition to the usual pre-assessment instructions, Include instructions on use of:
	● Code of Practice for Assessors (Annex A)

Pre-assessment instructions for candidates	● Be punctual ● Observe dress code: smart casual ● Request for assessor to repeat questions if question is not clear or if candidate has difficulty understanding assessor's accent ● To specify special needs, if any
Process of conducting assessment	● Verification of candidate’s identity ● Ensure candidates are seated properly so that they will not interact with each other ● Encourage the candidate to seek clarification if any ● Keep in mind the time limit for the assessment
Managing limitations of evidence	● Use of Oral Clarification
Recording Assessment Results	● Record the results in the Assessment Summary Record ● Check that all record forms, timing, dates, names, NRIC, signatures are completed. ● Check that all Assessment Criteria are ticked competent ● (C) or not yet competent (NYC) in the specific questions. As long as there is a NYC, the candidate is considered not successful in the entire assessment. The candidate is required to re-take the entire assessment. ● Record observation of Competence where applicable in ● the ‘Evidence of C and NYC must be recorded column ● Record reasons of Not Yet Competence where applicable in the ‘Evidence of C and NYC must be recorded column ● An ‘NYC’ in any K or A will results in an ‘NYC’ in the unit. ● A candidate is only deemed competent when they achieved a ‘C’ for all the A and K in the unit.
Feedback	● Give clear and constructive feedback to candidate using appropriate language ● Seek clarification and agreement with the candidate on the outcome ● Provide guidance on overcoming gaps in competency ● Ask the candidates to sign the assessment record ● Record the feedback on the assessment record ● Sign off the assessment record ● Assessor should reiterate the appeal process (if the candidate is assessed as not yet competent)

5.3.3. Assessment Specification for Oral Questioning (OQ)

Specifications	Oral Questioning (OQ)
Abilities and Knowledge	K4 A2
Duration	10 mins
Venue	Training room / designated assessment area
Set up	**Physical Assessment (Main)** Candidates are required to sit in a manner whereby there is sufficient separation so that they will not be able to interact during the assessment. **Virtual Assessment** LMS ● Questions to be pre-loaded into LMS and opened only to the candidates and during the assessment time. ● Alternatively, the assessor will send the questions electronically to candidates only during the assessment time. Virtual Environment ● Learners to ensure the camera is on all the time and showing their full face. ● Learners to share their whole screen throughout the assessment period Assessors to ensure the recording is started before the candidates and remained recordings for every breakout room throughout all the duration of the Assessment.

Pre-assessment instructions for assessors	● Put the candidate at ease
	● Explain clearly the purpose and context of the assessment:
	  ● Purpose of the assessment
	  ● Assessment duration
	  ● Assessment methods and tools
	● Explain the methods to be used, the standards and criteria for the assessment and the duration of the assessment.
	● Explain the procedures of the assessment:
	  ● Check whether the candidate has any special requirements e.g. language, special needs, etc.
	  ● Ensure that the following documents are made available
	  ● Assessment Summary Record
	  ● Candidate's Appeal Form
	● Brief candidate on his/her rights and process of appeal
	● Identify any allowable adjustments that candidates may require should they have a special need
	
	In addition to the usual pre-assessment instructions, Include instructions on use of:
	● Code of Practice for Assessors (Annex A)

Pre-assessment instructions for candidates	● Be punctual ● Observe dress code: smart casual ● Request for assessor to repeat questions if question is not clear or if candidate has difficulty understanding assessor's accent ● To specify special needs, if any
Process of conducting assessment	● Verification of candidate's identity ● Ensure candidates are seated properly so that they will not interact with each other ● Encourage the candidate to seek clarification if any ● Keep in mind the time limit for the assessment
Managing limitations of evidence	● Use of Oral Clarification
Recording Assessment Results	● Record the results in the Assessment Summary Record ● Check that all record forms, timing, dates, names, NRIC, signatures are completed. ● Check that all Assessment Criteria are ticked competent ● (C) or not yet competent (NYC) in the specific questions. As long as there is a NYC, the candidate is considered not successful in the entire assessment. The candidate is required to re-take the entire assessment. ● Record observation of Competence where applicable in ● the 'Evidence of C and NYC must be recorded column ● Record reasons of Not Yet Competence where applicable in the 'Evidence of C and NYC must be recorded column ● An 'NYC' in any K or A will results in an 'NYC' in the unit. ● A candidate is only deemed competent when they achieved a 'C' for all the A and K in the unit.
Feedback	● Give clear and constructive feedback to candidate using appropriate language ● Seek clarification and agreement with the candidate on the outcome ● Provide guidance on overcoming gaps in competency ● Ask the candidates to sign the assessment record ● Record the feedback on the assessment record ● Sign off the assessment record ● Assessor should reiterate the appeal process (if the candidate is assessed as not yet competent)

6. Assessment Records

6.1. Assessment Record for Written Assessment (SAQ)

Learning Outcomes	TSC Abilities and Knowledge	Tick		Evidence of “C” and “NYC” must be recorded
		C	NYC	
LO2: Understand and code R data types	K3 - Range and application of various types of data models K6 - Coding languages for programming of algorithms and signals K7 - Potential reasons for unintended outcomes A5 - Perform coding and configuration of software agents or programs based on a selected model or algorithm A9 - Implement changes to the coding and configuration of software			
LO3: Understand and use R packages and datasets	K3 - Range and application of various types of data models K6 - Coding languages for programming of algorithms and signals A2 - Use appropriate analytics platforms and analytical tools given specific analytics and reporting requirements A5 - Perform coding and configuration of software agents or programs based on a selected model or algorithm			

	A9 - Implement changes to the coding and configuration of software			
LO4: Perform data visualization in R	K3 - Range and application of various types of data models K5 - Statistical modelling techniques A3 - Utilize a range of statistical methods and analytics approaches to data A4 - Conduct statistical modelling of data to derive patterns / solutions A7 - Diagnose unintended outcomes produced by analytical models A10 -Draw relevant trends and insights from data analysis to support decisions			
LO5: Understand and code R programming language	K6 - Coding languages for programming of algorithms and signals K7 - Potential reasons for unintended outcomes A5 - Perform coding and configuration of software agents or programs based on a selected model or algorithm A7 - Diagnose unintended outcomes produced by analytical models A9 - Implement changes to the coding and configuration of software			
LO6: Understand and use statistical methods in R	K1 - Types of algorithms and advanced computational methods K2 - Range and application of various statistical algorithms K5 - Statistical modelling techniques			

	A1 - Identify appropriate statistical algorithms and data models to test hypotheses or theories A3 - Utilise a range of statistical methods and analytics approaches to data A4 - Conduct statistical modelling of data to derive patterns / solutions A6 - Conduct tests on the actions taken and outcomes to assess effectiveness of the model A7 - Diagnose unintended outcomes produced by analytical models			

C: Competent; NYC: Not Yet Competent

6.2. Assessment Record for Case Study

Enabling Learning Outcomes	TSC Abilities and Knowledge	Tick		Evidence of "C" and "NYC" must be recorded
		C	NYC	
LO2: Understand and code R data types	K3 - Range and application of various types of data models K6 - Coding languages for programming of algorithms and signals K7 - Potential reasons for unintended outcomes A5 - Perform coding and configuration of software agents or programs based on a selected model or algorithm A9 - Implement changes to the coding and configuration of software			
LO3: Understand and use R packages and datasets	K3 - Range and application of various types of data models K6 - Coding languages for programming of algorithms and signals A2 - Use appropriate analytics platforms and analytical tools given specific analytics and reporting requirements A5 - Perform coding and configuration of software agents or programs based on a selected model or algorithm			

	A9 - Implement changes to the coding and configuration of software			
LO4: Perform data visualization in R	K3 - Range and application of various types of data models K5 - Statistical modelling techniques A3 - Utilize a range of statistical methods and analytics approaches to data A4 - Conduct statistical modelling of data to derive patterns / solutions A7 - Diagnose unintended outcomes produced by analytical models A10 -Draw relevant trends and insights from data analysis to support decisions			
LO5: Understand and code R programming language	K6 - Coding languages for programming of algorithms and signals K7 - Potential reasons for unintended outcomes A5 - Perform coding and configuration of software agents or programs based on a selected model or algorithm A7 - Diagnose unintended outcomes produced by analytical models A9 - Implement changes to the coding and configuration of software			

LO6: Understand and use statistical methods in R	K1 - Types of algorithms and advanced computational methods K2 - Range and application of various statistical algorithms K5 - Statistical modelling techniques A1 - Identify appropriate statistical algorithms and data models to test hypotheses or theories A3 - Utilise a range of statistical methods and analytics approaches to data A4 - Conduct statistical modelling of data to derive patterns / solutions A6 - Conduct tests on the actions taken and outcomes to assess effectiveness of the model A7 - Diagnose unintended outcomes produced by analytical models			

C: Competent; NYC: Not Yet Competent

6.3. Assessment Record for Oral Questioning (OQ)

Enabling Learning Outcomes	TSC Abilities and Knowledge	Tick		Evidence of "C" and "NYC" must be recorded
		C	NYC	
LO1: Install R and R Studio IDE	K4 - Usage of analytics platforms and tools A2 - Use appropriate analytics platforms and analytical tools given specific analytics and reporting requirements			

C: Competent; NYC: Not Yet Competent

6.4. Assessment Summary Record

Learning Outcomes	Abilities and Knowledge	Assessment Methods			Overall Result Indicate C or NYC
		WA (SAQ)	CS	OQ	
LO1: Install R and R Studio IDE	K4 - Usage of analytics platforms and tools A2 - Use appropriate analytics platforms and analytical tools given specific analytics and reporting requirements	-	-	√	
LO2: Understand and code R data types	K3 - Range and application of various types of data models K6 - Coding languages for programming of algorithms and signals K7 - Potential reasons for unintended outcomes A5 - Perform coding and configuration of software agents or programs based on a selected model or algorithm A9 - Implement changes to the coding and configuration of software	√	√	-	
LO3: Understand and use R packages and datasets	K3 - Range and application of various types of data models K6 - Coding languages for programming of algorithms and signals A2 - Use appropriate analytics platforms and analytical tools given specific analytics and reporting requirements A5 - Perform coding and configuration of software agents or programs based on a selected model or algorithm A9 - Implement changes to the coding and configuration of software	√	√	-	

Learning Outcomes	Abilities and Knowledge	Assessment Methods			Overall Result Indicate
		WA (SAQ)	CS	OQ	C or NYC
LO4: Perform data visualization in R	K3 - Range and application of various types of data models K5 - Statistical modelling techniques A3 - Utilise a range of statistical methods and analytics approaches to data A4 - Conduct statistical modelling of data to derive patterns / solutions A7 - Diagnose unintended outcomes produced by analytical models A10 -Draw relevant trends and insights from data analysis to support decisions	√	√	-	
LO5: Understand and code R programming language	K6 - Coding languages for programming of algorithms and signals K7 - Potential reasons for unintended outcomes A5 - Perform coding and configuration of software agents or programs based on a selected model or algorithm A7 - Diagnose unintended outcomes produced by analytical models A9 - Implement changes to the coding and configuration of software	√	√	-	
LO6: Understand and use statistical methods in R	K1 - Types of algorithms and advanced computational methods K2 - Range and application of various statistical algorithms K5 - Statistical modelling techniques	√	√	-	

Learning Outcomes	Abilities and Knowledge	Assessment Methods			Overall Result Indicate
		WA (SAQ)	CS	OQ	C or NYC
	A1 - Identify appropriate statistical algorithms and data models to test hypotheses or theories A3 - Utilise a range of statistical methods and analytics approaches to data A4 - Conduct statistical modelling of data to derive patterns / solutions A6 - Conduct tests on the actions taken and outcomes to assess effectiveness of the model A7 - Diagnose unintended outcomes produced by analytical models				

DE: Direct Evidence - Observation of performance and authentic work products; WA: Written Assessment; CS: Case Study; OQ: Oral Questioning

This candidate has been assessed as

☐ **COMPETENT**

☐ **NOT YET COMPETENT**

Candidate Name (As in NRIC)		Assessor Name	
NRIC (Last 3 digits & alphabet)		NRIC (Last 3 digits & alphabet)	
Candidate Signature		Assessor Signature	
Date:		Date:	

By signing, the candidate is agreeing to accept the assessment outcome	***By signing, the assessor is agreeing to have duly assessed the performance criteria and underpinning knowledge as required in the competence units***

Feedback on outcome by Assessor/ Feedback by candidate:
{Feedback on the overall performance or in the case of NYC; any area of skills gap and improvement needed}

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7. Annexes

7.1. Annex A: Code of Practice for Assessors

1) The differing needs and requirements of the person(s) being assessed, the local enterprise(s) and/or industry are identified and handled with sensitivity.

2) Potential forms of conflict of interest in the assessment process and/or outcomes are identified and appropriate referrals are made, if necessary.

3) All forms of harassment are avoided throughout the planning, conduct, reviewing and reporting of the assessment outcomes.

4) The rights of the candidate(s) are protected during and after the assessment.

5) Personal or interpersonal factors that are not relevant to the assessment of competency must not influence the assessment outcomes.

6) The candidate(s) is made aware of rights and processes of appeal.

7) Evidence that is gathered during the assessment is verified for validity, reliability, authenticity, sufficiency and currency.

8) Assessment decisions are based on available evidence that can be produced and verified by another assessor.

9) Assessments are conducted within the boundaries of the assessment system policies and procedures.

10) Formal agreement is obtained from both the candidate(s) and the assessor that the assessment is carried out in accordance with agreed procedures.

11) Assessment tools, systems, and procedures are consistent with equal opportunity legislation.

12) The candidate(s) is informed of all assessment reporting processes prior to the assessment.

13) The candidate(s) is informed of all known potential consequences of decisions arising from an assessment, prior to the assessment.

14) Confidentiality is maintained regarding assessment results.

15) Results are only released with the written permission of the candidate(s).

16) The assessment results are used consistently with the purposes explained to the candidate.

17) Self-assessments are periodically conducted to ensure current competencies against the assessment and Workplace Training Competency Standards.

18) Professional development opportunities are identified and sought.

19) Opportunities for networking amongst assessors are created and maintained.

20) Opportunities are created for technical assistance in planning, conducting and reviewing assessment procedures and outcomes.

7.2. Annex B: Checklist for Assessors

S/No.	Items	Tick (√)
1.	Verify candidate’s identity	
2.	Explain the purpose, process and duration of assessment	
3.	Explain the appeal process	
4.	Check with candidate for any special needs	
5.	Ask candidate to clarify doubts	
6.	Check if candidate is ready to proceed with assessment	
7.	Record the start and end time of assessment	
8.	Record all evidence and comments/feedback in the Assessment Record	
9.	Provide feedback to candidate at the end of assessment	
10.	Record results and feedback in Summary Assessment Record *(A candidate is deemed competent when she/he achieved a “C” for all abilities and knowledge.)*	
11.	Sign appropriate pages	
12.	Ensure candidate signs on Summary Assessment Record	
13.	Submit all completed forms to the Person in Charge	

7.3. Annex C: Pre-Assessment Information for Candidate

Assessors are required to provide candidates with relevant pre-assessment information which includes, but is not limited

1) This assessment of a range of assessment methods, including Written Assignment (Q&A), Written Assignment (Case Study) and Oral Questioning. The conduct of assessment may be one-on-one, and, in a group, context as outlined in the Assessment Specifications.
2) For these competency elements and the required performance criteria (including range and context) will be assessed. Refer to the Assessment Specifications and 'Instructions for Candidates' for a detailed coverage of the assessment requirements.
3) You have a right of appeal to assessment decisions in accordance with Tertiary Infotech Pte Ltd appeal procedures
4) The conditions for granting a deferred assessment are:
    - Medical grounds – a medical report or certificate from registered medical practitioners
    - Unexpected and exceptional circumstances which may include (but are not limited to):
    - accidents (sporting, motor vehicle, etc.) where an injury is sustained bereavement

Unexpected and exceptional circumstances ***does not*** include cases where candidates have mistaken the day, time or venue of assessment.

7.4. Annex D: Assessment Questions and Answers

7.4.1. Assessment Questions and Answers for WA(SAQ)

Refer to the document

- Written Assessment (SAQ)
- Answers to Written Assessment (SAQ)

7.4.2. Assessment Questions and Case Study

Refer to the document

- Case Study Assessment (CS)
- Answers to CS Assessment

7.4.3. Assessment Questions and Oral Questioning (OQ)

Refer to the document

- Oral Questioning (OQ)
- Answers to Oral Questioning (OQ)